Cricket Data Scraping

A Data-Driven Analysis Of IPL Team Performance And Sustained Rankings (2022–2024)

LEKHANA M S

Cricket Data Scraping

Abstract

This study presents a multi-season analytical evaluation of Indian Premier League (IPL) team performance across the 2022, 2023, and 2024 seasons using data-driven performance indicators. Match-level points table data were collected from official IPL sources and processed through systematic data cleaning and normalization techniques to ensure consistency and comparability across seasons. Key performance metrics, including Win Percentage, Net Run Rate (NRR), Batting Strength, Bowling Strength, and Balanced Performance Index, were derived to capture both outcome-based and skill-based dimensions of team performance.

A composite Season Dominance Index (SDI) was developed using weighted normalized metrics to quantify overall team dominance within each season. To assess long-term performance sustainability, SDI values were averaged across the three seasons to generate overall rankings, emphasizing consistency rather than single-season success. Additional analysis using a Consistency Index highlighted variations in performance stability among teams.

The results indicate that teams such as Rajasthan Royals and Kolkata Knight Riders demonstrate sustained dominance and balanced performance across seasons, while others exhibit performance volatility or reliance on isolated strengths. The study underscores the importance of combining result-oriented metrics with underlying performance indicators to achieve a holistic and reliable assessment of long-term team competitiveness in the IPL.

Introduction

In competitive sports analysis, evaluating team performance solely based on match results often provides an incomplete understanding of overall team strength. While wins and losses indicate outcomes, they do not fully capture performance quality, balance across skill areas, or consistency over time. This limitation is particularly evident in a tournament such as the Indian Premier League (IPL), where team performances can vary significantly across seasons.

The primary aim of this study is to identify sustained performance rankings of IPL teams across the 2022, 2023, and 2024 seasons. To support this objective, team rankings were generated separately for each season across multiple performance metrics, including Win Percentage, Net Run Rate (NRR), Batting Strength, Bowling Strength, Balanced Performance Index, and overall performance measures. These season-wise rankings enable a detailed evaluation of how teams perform under different criteria within individual seasons.

In addition to conventional metrics, this study introduces two newly developed indices: the Season Dominance Index (SDI) and the Consistency Index. The SDI is a composite measure designed to quantify overall team dominance within a season by integrating results-based and performance-based indicators. The Consistency Index, on the other hand, assesses the

stability of team performance across multiple seasons, providing insight into long-term reliability rather than short-term success.

To enhance interpretability and facilitate clearer comparison, data visualisation techniques were employed throughout the analysis. These visual representations help identify performance patterns, variations across metrics, and the relationship between dominance and consistency, thereby supporting a comprehensive and objective assessment of sustained team performance in the IPL**.**

Process Followed

1. Data Collection and Initial Feature Setup

Data Source
Season-wise IPL team-level points table data for the 2022, 2023, and 2024 seasons was collected from the official IPL T20 website.

The tabular data was extracted directly into Microsoft Excel using a Chrome web scraping extension, ensuring accurate, efficient, and repeatable data collection.

2. Data Cleaning and Pre-Processing

The raw scraped data was processed using advanced Excel operations to address inconsistencies and prepare it for analysis.

Key cleaning steps included:

- Removal of inconsistencies and formatting issues
- Handling missing values by identifying blanks and filling them appropriately
- Standardization of data formats across all seasons

Excel techniques used:

- Find and Replace
- Text functions: TRIM, UPPER, LOWER, PROPER, SUBSTITUTE, TEXT
- Remove Duplicates
- Handling missing data (filling blanks)
- Conditional Formatting for error identification

Additional corrections performed:

- Corrected formatting and sign inconsistencies in Net Run Rate (NRR)
- Standardized team names to ensure consistency across seasons

- Converted all numeric fields to appropriate data types
- Removed duplicate records
- Split combined columns where required and merged season-wise datasets into a unified structure

These steps ensured that the dataset was clean, consistent, and analysis-ready for subsequent feature engineering and index calculations.

Team-Level Data Description

The following variables were collected from the IPL points table and used for performance analysis. Each original field was renamed for clarity and consistency.

Original Column	Suggested Name	Description
POS	Rank	Final position of the team in the season points table
TEAM	Team Name	Name of the IPL team
P	Matches Played	Total number of matches played by the team
W	Wins	Total number of matches won
L	Losses	Total number of matches lost
NR	No Result	Number of matches with no result due to rain or abandonment
NRR	Net Run Rate (NRR)	Net run rate, indicating the scoring and conceding efficiency of the team
FOR	Total Runs Scored	Total runs scored by the team across the tournament
AGAINST	Total Runs Conceded	Total runs conceded by the team across the tournament
PTS	Points	Total points earned (2 points awarded per win)

Derived Performance Metrics

To enable fair and meaningful comparison across teams and seasons, several additional performance metrics were derived from the base dataset.

---

1. Win Percentage

Win Percentage was calculated to measure match-winning efficiency independent of the number of matches played.

$$\text{Win Percentage} = \left(\frac{\text{Wins}}{\text{Matches Played}}\right) \times 100$$

This metric reflects a team's ability to convert matches into wins, regardless of season length.

---

2. Batting Runs per Over (Batting RPO)

Batting efficiency was evaluated using runs scored per over.

$$\text{Batting RPO} = \frac{\text{Total Runs Scored}}{\text{Total Overs Faced}}$$

Higher values indicate stronger and more efficient batting performance across the season.

---

3. Bowling Runs per Over (Bowling RPO)

Bowling effectiveness was measured using runs conceded per over.

$$\text{Bowling RPO} = \frac{\text{Total Runs Conceded}}{\text{Total Overs Bowled}}$$

Lower values represent stronger bowling performance.

---

4. Batting and Bowling Strength

Since raw RPO values are not directly comparable across teams, both batting and bowling metrics were normalized using min–max scaling to create standardized strength indices.

Batting Strength

$$\text{Batting Strength} = \frac{\text{Batting RPO} - \text{Min Batting RPO}}{\text{Max Batting RPO} - \text{Min Batting RPO}} \times 100$$

Higher values indicate stronger batting performance.

Bowling Strength

Bowling RPO was normalized using an inverted min–max approach so that lower run concession corresponds to higher strength.

$$\text{Bowling Strength} = \frac{\text{Max Bowling RPO} - \text{Bowling RPO}}{\text{Max Bowling RPO} - \text{Min Bowling RPO}} \times 100$$

This ensures a consistent interpretation across strength metrics, where higher values represent better performance.

---

5. Normalized Net Run Rate (NRR)

Net Run Rate values vary across teams and seasons. To ensure comparability, NRR was normalized using min–max scaling.

$$\text{NRR}_{\text{norm}} = \frac{\text{NRR} - \text{Min NRR}}{\text{Max NRR} - \text{Min NRR}} \times 100$$

Higher values indicate better overall run rate performance.

---

6. Season Dominance Index (SDI)

The **Season Dominance Index (SDI)** is a composite metric designed to quantify a team's overall dominance within a season. It combines match outcomes with performance efficiency, capturing both results and underlying team quality.

SDI Formula

$$\text{SDI} = 0.40 \times \text{Win Percentage}_{\text{norm}} + 0.25 \times \text{NRR}_{\text{norm}} + 0.20 \times \text{Batting Strength} + 0.15 \times \text{Bowling Strength}$$

Metric	**Weight**
Win Percentage	40%
Normalized NRR	25%
Batting Strength	20%
Bowling Strength	15%
Total	**100%**

Interpretation of SDI

- **Higher SDI (closer to 100):** Indicates a dominant team with strong match outcomes, high efficiency in batting and bowling, and balanced overall performance.
- **Lower SDI (closer to 0):** Indicates lower dominance, resulting from weaker results, lower efficiency, or imbalanced performance across skill areas.

7. Balanced Performance Index (BPI)

The **Balanced Performance Index (BPI)** measures a team's overall all-round capability by combining batting and bowling strengths.

$$\text{BPI} = \frac{\text{Batting Strength} + \text{Bowling Strength}}{2}$$

Interpretation:

- Higher BPI values indicate stronger all-round performance.

- Values range from **0 (weak overall performance)** to **100 (excellent all-round team)**.
- This index emphasizes balance, rewarding teams that perform well in both batting and bowling rather than excelling in only one discipline.

8. Team Rankings Across Metrics

After calculating the normalized metrics and indices for each season (2022, 2023, 2024), teams were ranked separately for each metric to facilitate comparison:

Metric	Ranking Method	Purpose
Win Percentage	Descending order	Measures match-winning efficiency
Normalized NRR	Descending order	Captures dominance through margins of wins and losses
Batting Strength	Descending order	Indicates batting efficiency
Bowling Strength	Descending order	Indicates bowling effectiveness
Balanced Performance Index (BPI)	Descending order	Highlights all-round team balance
Season Dominance Index (SDI)	Descending order	Identifies the most dominant team in a given season

- Ranking ensures comparability across seasons, independent of raw metric values.
- In cases of ties, secondary metrics such as NRR or BPI were used for resolution.

SDI Rankings Interpretation

The SDI rankings provide a holistic measure of team dominance within a season.
Teams with higher SDI ranks consistently combine match success, performance efficiency, and balanced skill execution, offering a more accurate reflection of seasonal dominance than win-based rankings alone.

2024 Rankings

Win Percentage Ranks			NRR Ranks			Batting Strength		
1	KKR	64.29%	1	KKR	100.00%	1	KKR	100.00%
2	SRH	57.14%	2	RCB	61.14%	2	SRH	96.53%
3	RR	57.14%	3	SRH	59.12%	3	RCB	82.23%
4	RCB	50.00%	4	CSK	58.63%	4	DC	52.23%
5	CSK	50.00%	5	RR	53.78%	5	MI	48.70%
6	DC	50.00%	6	MI	30.20%	6	RR	33.80%
7	LSG	50.00%	7	PBKS	28.62%	7	CSK	30.87%
8	GT	35.71%	8	DC	27.17%	8	PBKS	24.47%
9	PBKS	35.71%	9	LSG	16.13%	9	LSG	18.68%
10	MI	28.57%	10	GT	0.00%	10	GT	0.00%
Bowling Strength			**Season Dominance Index**			**Balanced Performer Index**		
1	CSK	100.00%	1	KKR	81.33%	1	KKR	85.39%
2	RR	86.48%	2	SRH	56.94%	2	CSK	65.44%
3	KKR	70.78%	3	RR	56.03%	3	RR	60.14%
4	PBKS	53.96%	4	CSK	55.83%	4	RCB	53.99%
5	LSG	39.61%	5	RCB	55.59%	5	SRH	48.27%
6	GT	38.29%	6	DC	38.53%	6	PBKS	39.22%
7	RCB	25.75%	7	PBKS	34.43%	7	MI	34.19%
8	MI	19.68%	8	LSG	33.71%	8	DC	30.42%
9	DC	8.61%	9	MI	31.67%	9	LSG	29.14%
10	SRH	0.00%	10	GT	20.03%	10	GT	19.15%

2023 Rankings

Win Percentage Ranks			NRR Ranks			Batting Strength		
1	GT	71.43%	1	GT	100.00%	1	MI	100.00%
2	CSK	57.14%	2	RR	90.13%	2	CSK	83.75%
3	LSG	57.14%	3	LSG	67.54%	3	PBKS	73.69%
4	MI	57.14%	4	DC	59.09%	4	GT	73.21%
5	RR	50.00%	5	KKR	58.71%	5	RCB	70.04%
6	RCB	50.00%	6	PBKS	47.66%	6	KKR	63.76%
7	KKR	42.86%	7	RCB	35.70%	7	RR	58.47%
8	PBKS	42.86%	8	SRH	31.28%	8	LSG	54.80%
9	DC	35.71%	9	CSK	13.50%	9	SRH	39.72%
10	SRH	28.57%	10	MI	0.00%	10	DC	0.00%
Bowling Strength			**Season Dominance Index**			**Balanced Performer Index**		
1	GT	100.00%	1	GT	83.21%	1	GT	86.61%
2	LSG	83.42%	2	CSK	73.25%	2	CSK	78.89%
3	CSK	74.03%	3	LSG	63.22%	3	LSG	69.11%
4	DC	70.31%	4	RR	56.69%	4	RR	63.30%
5	RR	68.14%	5	RCB	56.58%	5	RCB	61.34%
6	RCB	52.64%	6	MI	54.77%	6	MI	50.00%
7	SRH	35.61%	7	KKR	43.64%	7	KKR	47.93%
8	KKR	32.11%	8	PBKS	41.76%	8	PBKS	43.70%
9	PBKS	13.72%	9	SRH	28.09%	9	SRH	37.66%
10	MI	0.00%	10	DC	24.83%	10	DC	35.15%

2022 Rankings

Win Percentage Ranks			NRR Ranks			Batting Strength		
1	GT	71.43%	1	GT	100.00%	1	RR	100.00%
2	SRH	64.29%	2	RR	98.93%	2	DC	96.56%
3	RR	64.29%	3	LSG	95.94%	3	PBKS	78.79%
4	RCB	57.14%	4	DC	92.63%	4	LSG	45.46%
5	CSK	50.00%	5	KKR	91.71%	5	GT	41.63%
6	DC	50.00%	6	PBKS	87.96%	6	SRH	32.26%
7	LSG	42.86%	7	RCB	63.92%	7	KKR	29.61%
8	GT	42.86%	8	SRH	51.88%	8	RCB	16.95%
9	PBKS	28.57%	9	CSK	2.25%	9	CSK	7.95%
10	MI	28.57%	10	MI	0.00%	10	MI	0.00%
Bowling Strength			**Season Dominance Index**			**Balanced Performer Index**		
1	GT	100.00%	1	RR	80.38%	1	RR	83.12%
2	KKR	96.62%	2	GT	76.90%	2	DC	78.53%
3	LSG	92.94%	3	LSG	72.73%	3	PBKS	71.75%
4	RCB	69.91%	4	DC	71.54%	4	GT	70.82%
5	RR	66.23%	5	PBKS	67.46%	5	LSG	69.20%
6	PBKS	64.71%	6	KKR	60.48%	6	KKR	63.11%
7	DC	60.50%	7	RCB	52.71%	7	RCB	43.43%
8	SRH	46.76%	8	SRH	43.58%	8	SRH	39.51%
9	MI	1.69%	9	CSK	13.58%	9	CSK	3.98%
10	CSK	0.00%	10	MI	11.68%	10	MI	0.85%

9. Overall rankings

The overall rankings were obtained by averaging each team's performance metrics across the 2022, 2023, and 2024 seasons. These averaged values were then used to create a final consolidated ranking, ensuring that the results reflect consistent performance over time rather than short-term success in a single season.

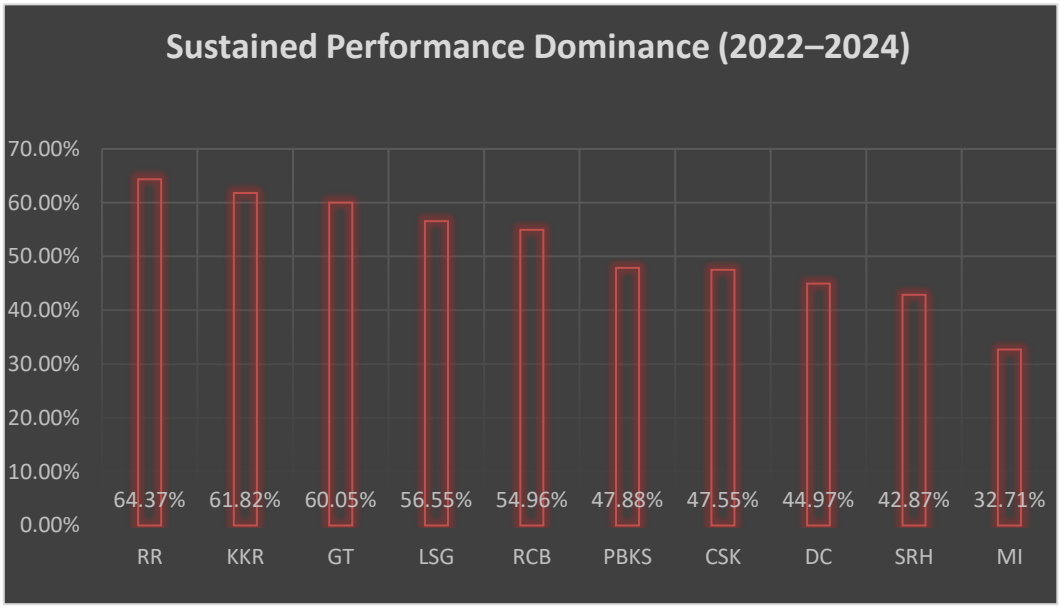
Win Percentage Ranks			NRR Ranks			Batting Strength		
1	GT	59.52%	1	KKR	75.80%	1	KKR	64.46%
2	LSG	57.14%	2	RR	70.60%	2	RR	64.09%
3	RR	57.14%	3	GT	66.67%	3	PBKS	58.98%
4	RCB	52.38%	4	RCB	61.26%	4	RCB	56.40%
5	KKR	50.00%	5	LSG	59.87%	5	SRH	56.17%
6	DC	45.24%	6	CSK	50.34%	6	DC	49.59%
7	CSK	45.24%	7	PBKS	49.29%	7	MI	49.57%
8	PBKS	42.86%	8	SRH	41.50%	8	CSK	40.86%
9	SRH	42.86%	9	DC	39.93%	9	LSG	39.65%
10	MI	38.10%	10	MI	25.95%	10	GT	38.28%
Bowling Strength			**Season Dominance Index**			**Balanced Performer Index**		
1	GT	79.43%	1	RR	64.37%	1	RR	68.85%
2	RR	73.62%	2	KKR	61.82%	2	KKR	65.48%
3	LSG	71.99%	3	GT	60.05%	3	GT	58.86%
4	KKR	66.50%	4	LSG	56.55%	4	LSG	55.82%
5	CSK	58.01%	5	RCB	54.96%	5	RCB	52.92%
6	RCB	49.43%	6	PBKS	47.88%	6	PBKS	51.56%
7	DC	46.47%	7	CSK	47.55%	7	CSK	49.44%
8	PBKS	44.13%	8	DC	44.97%	8	DC	48.03%
9	SRH	27.46%	9	SRH	42.87%	9	SRH	41.81%
10	MI	7.12%	10	MI	32.71%	10	MI	28.35%

10. Overall Dominance Measurement Using Average SDI

To evaluate overall team performance across multiple seasons, the Season Dominance Index (SDI) values for each team were averaged over the three seasons from 2022 to 2024. This Average SDI was used as the final metric for the Sustained Performance Rankings (2022–2024).

Unlike single-season rankings, this approach captures not only peak performance in an individual year but also a team's ability to remain competitive and perform consistently over time. By using the average SDI as the final ranking measure, the analysis presents a balanced assessment that reflects match results, performance efficiency, and long-term consistency.

The graph below illustrates the final team rankings based on the average Season Dominance Index (SDI) across the 2022–2024 seasons, highlighting teams that have demonstrated sustained dominance in the league.



The chart highlights **sustained performance dominance** across the **2022–2024 seasons**. **Rajasthan Royals (RR)** and **Kolkata Knight Riders (KKR)** lead the rankings, indicating consistently high-level performance over multiple years. Teams such as **Gujarat Titans (GT)**, **Lucknow Super Giants (LSG)**, and **Royal Challengers Bengaluru (RCB)** form a strong middle tier, demonstrating stable performance but slightly lower overall dominance. In contrast, **Mumbai Indians (MI)** and **Sunrisers Hyderabad (SRH)** appear at the lower end of the rankings, reflecting comparatively inconsistent performance across the three seasons.

---

11. Consistency Index

In addition to average performance, the **standard deviation of SDI** was calculated for each team to capture fluctuations in seasonal dominance. This variability was then normalised to compute a **Deviation Percentage**, allowing fair comparison across teams. To translate this variability into an intuitive measure of consistency, a **Consistency Index (CI)** was derived using the following normalisation formula:

$$\text{Consistency Index} = 1 - \frac{D - \min(D)}{\max(D) - \min(D)}$$

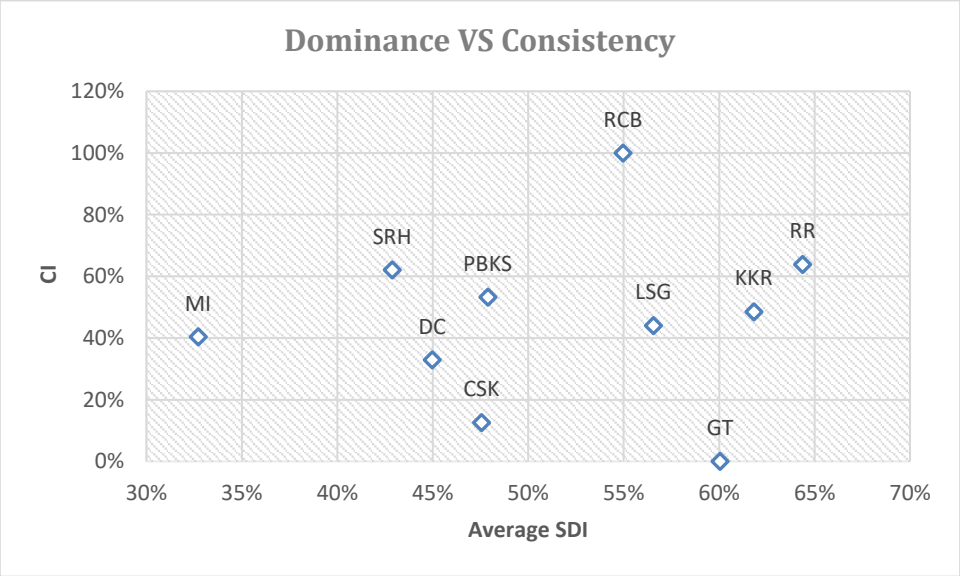
where **D** represents the standard deviation of SDI for a team across seasons.

This transformation ensures that teams with **lower variation in SDI values (higher consistency)** receive higher consistency scores, while teams with **greater fluctuations in performance** are assigned lower consistency values.

All Rankings Comparison								
SL. NO	**Teams**	**Win % Rank**	**NRR Rank**	**Batting Strength Rank**	**Bowling Strength Rank**	**Balanced Performance**	**Overall Performance Rank**	**Consistancy Rank**
1	GT	1	3	10	1	3	3	10
2	RR	3	2	2	2	1	1	2
3	LSG	2	5	9	3	4	4	6
4	RCB	4	4	4	6	5	5	1
5	DC	6	9	6	7	8	8	8
6	PBKS	8	7	3	8	6	6	4
7	KKR	5	1	1	4	2	2	5
8	SRH	9	8	5	9	9	9	3
9	CSK	7	6	8	5	7	7	9
10	MI	10	10	7	10	10	10	7

12. Data visualisation And Insights

a. **Consistency vs Dominance scatter plot**, plotting average SDI against the Consistency Index to assess overall performance and stability across seasons.



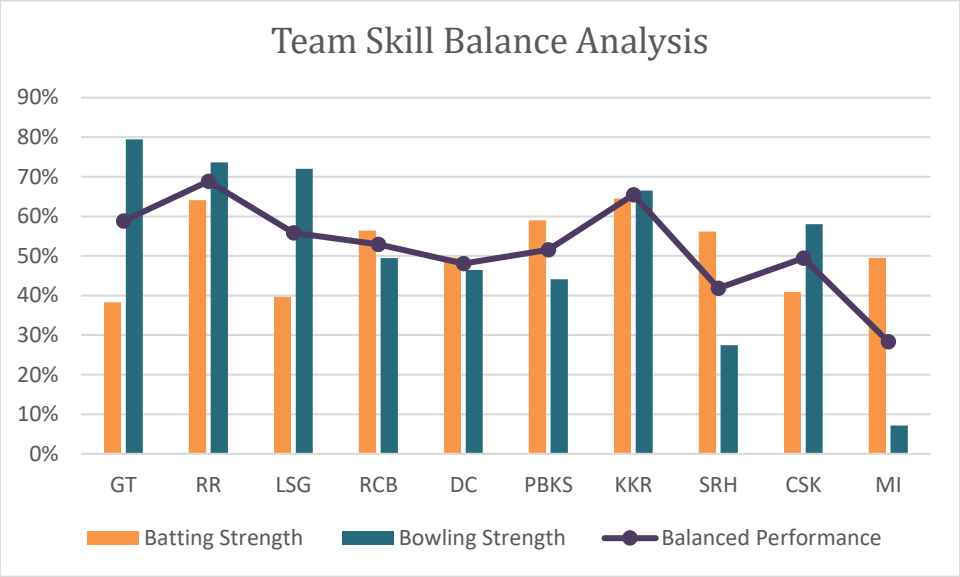
- RCB and RR emerge as the most consistent high-performing teams. RCB stands out for its exceptional consistency despite having moderate overall dominance, while RR successfully combines high dominance with strong consistency across seasons.
- GT and MI show contrasting performance patterns. GT displays high dominance but lower consistency, suggesting fluctuations across seasons, whereas MI records relatively

better consistency despite lower dominance, indicating a more stable—though less dominant—performance profile**.**

-

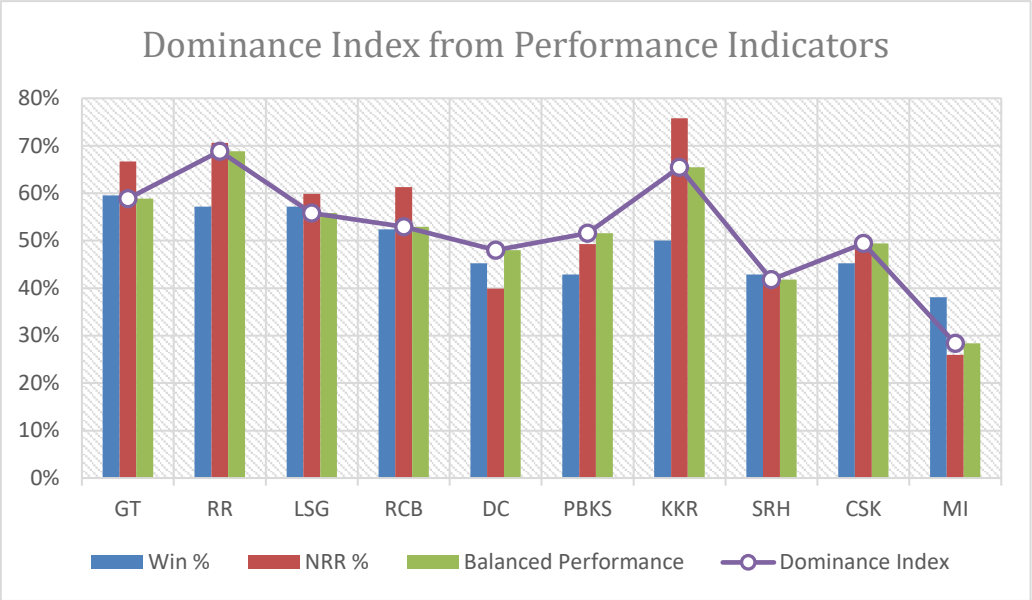
b. Team Skill Balance Analysis

This analysis compares teams based on their **batting strength, bowling strength, and the resulting balanced performance index**. The objective is to identify how evenly teams perform across core skill areas and to highlight whether overall performance is driven by balance or by reliance on a single strength.



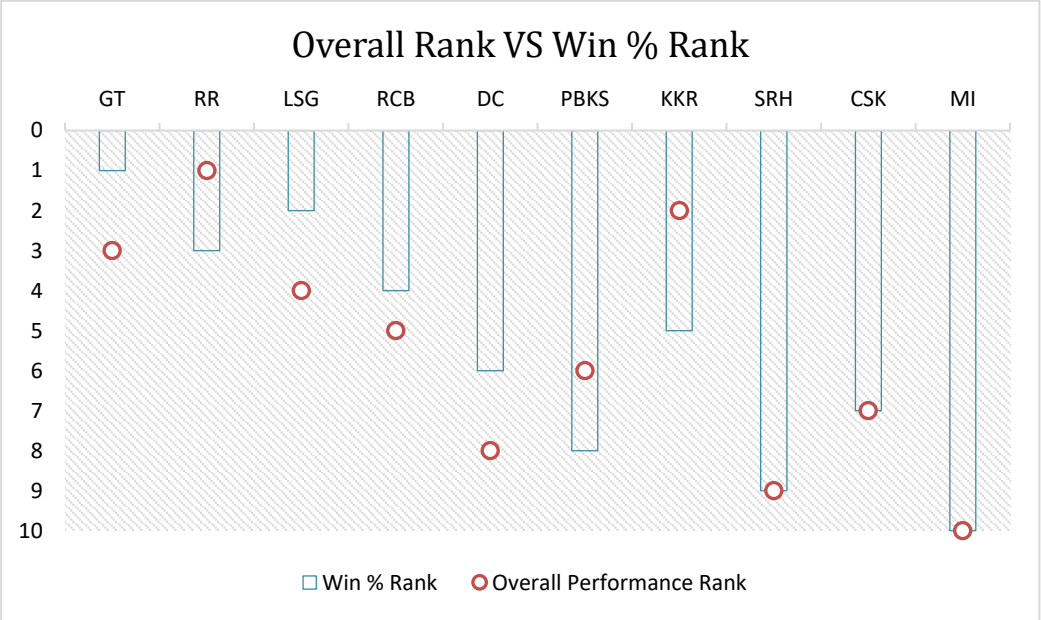
- RR and KKR display the strongest overall balance, with batting and bowling strengths contributing evenly, leading to the highest balanced performance scores.
- GT and MI show clear skill asymmetry. GT's dominance is driven primarily by strong bowling performance, while MI's weaker bowling significantly lowers its overall balance despite having moderate batting strength.

c. Dominance Index Analysis Based on Performance Indicators



- KKR and RR show the highest dominance, reflecting strong and balanced performance across win percentage, NRR, and consistency.
- GT, LSG, and RCB demonstrate moderate dominance with relatively stable performance across indicators.
- MI records the lowest dominance index, indicating overall inconsistency across key performance indicators.

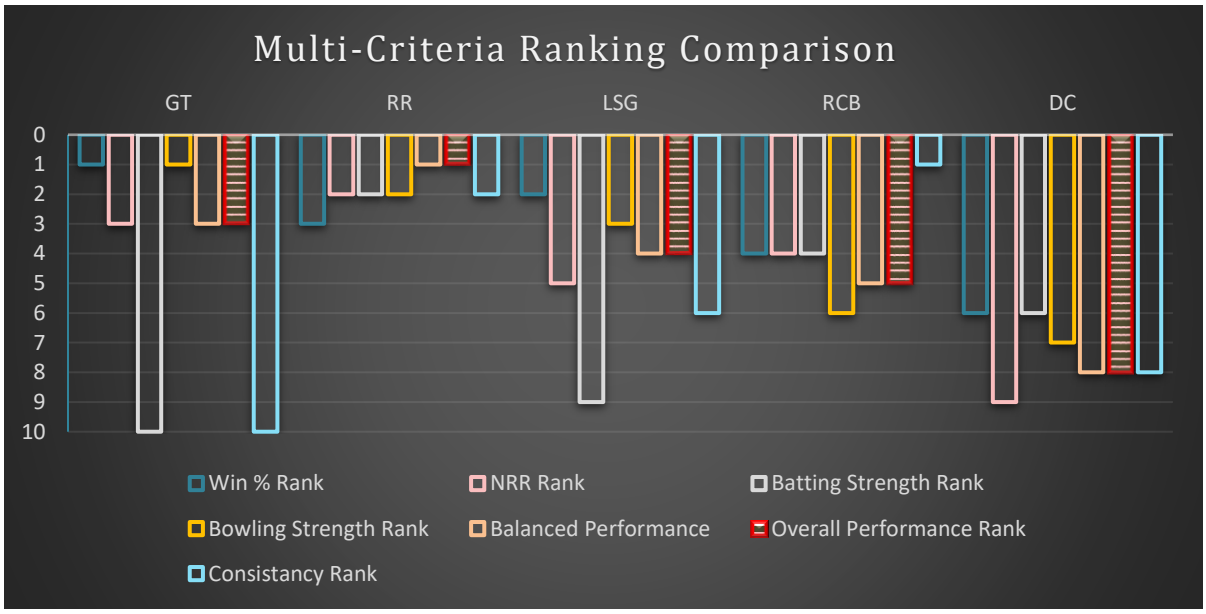
d. Overall Rank vs Win % Rank

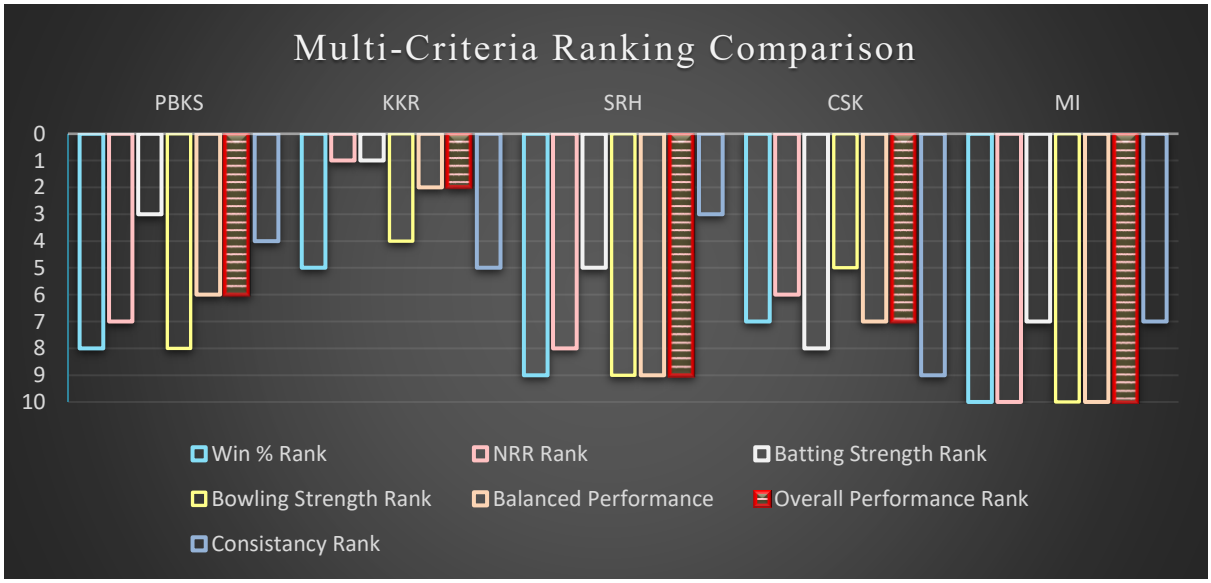


- This chart shows that overall performance rankings do not always match win percentage rankings. While some teams retain similar positions in both, others experience noticeable shifts due to the influence of Net Run Rate, batting strength, bowling strength, and consistency, highlighting that sustained dominance extends beyond match wins alone.

e. Multi-Criteria Ranking Comparison Across Teams

This chart compares team rankings across multiple performance criteria, including win percentage, Net Run Rate, batting strength, bowling strength, consistency, and overall performance, offering a comprehensive view of team performance. The visualization shows that overall rankings and consistency are shaped by a combination of results-based metrics (win percentage and NRR) and skill-based metrics (batting, bowling, and balance), rather than being driven by any single factor alone.





- **Top Teams:** Teams at the top of the rankings show consistently strong performance across win percentage, Net Run Rate, batting strength, bowling strength, and consistency. Their high SDI and overall rankings reflect a well-balanced skill set and sustained dominance, rather than dependence on any single performance metric.
- **Bottom Teams:** Teams placed lower in the rankings display noticeable variation across metrics, often performing adequately in one area but underperforming in others. This imbalance results in lower SDI and overall rankings, indicating a lack of sustained and well-rounded performance across seasons.

Key Findings

1. Top-Ranked Teams (Sustained Performers)

- **Rajasthan Royals (RR) and Kolkata Knight Riders (KKR)** emerge as the top sustained performers across the 2022–2024 seasons, driven by strong dominance, balanced batting and bowling, and high consistency.

- **Gujarat Titans (GT), Lucknow Super Giants (LSG), and Royal Challengers Bengaluru (RCB)** form a competitive middle tier, showing stable performance but slightly lower consistency or balance compared to the top teams.

- **Mumbai Indians (MI) and Sunrisers Hyderabad (SRH)** rank lower due to inconsistent performance and imbalances between batting and bowling.

2. Other Metrics Evaluation

- **Batting, Bowling, and Balance:** Rajasthan Royals (RR) emerge as the strongest batting team, Gujarat Titans (GT) stand out as the most effective bowling side, and Kolkata Knight Riders (KKR) demonstrate the best overall balance with evenly matched batting and bowling strengths, resulting in a high Balanced Performance Index and sustained all-round performance.

- **Consistency Evaluation :** Royal Challengers Bengaluru (RCB) and Rajasthan Royals (RR) emerge as the most consistent teams across the 2022–2024 seasons. RCB shows exceptional stability despite moderate dominance, while RR combines strong consistency with high overall dominance, making both teams reliable long-term performers.

- **Win Percentage:** Rajasthan Royals (RR) record the highest win percentage across the seasons, reflecting strong match-winning efficiency, while Sunrisers Hyderabad (SRH) register the lowest win percentage, indicating comparatively weaker results-based performance.

Conclusion

This analysis demonstrates that sustained success in the IPL cannot be judged by match wins alone. By combining win percentage, Net Run Rate, batting strength, bowling strength, balance performance and consistency into composite indices such as SDI & CI, the study provides a more accurate and holistic measure of team performance across the 2022–2024 seasons.

The results clearly show that teams with balanced skill profiles and consistent performance across seasons achieve higher long-term dominance. Rajasthan Royals and Kolkata Knight Riders stand out as the most well-rounded and reliable teams, reflecting both strong results and efficient on-field performance. In contrast, teams that rely heavily on a single strength—either batting or bowling—tend to experience fluctuations in rankings and lower overall dominance.

The Season Dominance Index and related metrics provide a robust framework for evaluating teams beyond traditional standings and offer valuable insights for performance analysis, strategic planning, and future team development.

Limitations

This study is based on team-level data collected from the official IPL points tables for the 2022, 2023, and 2024 seasons. As a result, individual player performances and match-level situational factors are not considered in the analysis. Additionally, the evaluation is limited to three seasons, which, while sufficient for identifying short- to medium-term trends, may not fully capture longer-term performance patterns. The derived indices, including the Season Dominance Index and Consistency Index, rely on selected performance metrics and assigned weightages, which may influence the final rankings. Changes in metric selection or weighting could lead to variations in results.

Future Scope

The scope of this analysis can be extended in several ways.

- The findings provide a strong foundation for **probability-based analysis**. The derived performance metrics and indices can be used to estimate the probability of match outcomes in future IPL tournaments. By extending the dataset to include additional seasons and integrating match-level and player-level data, more accurate probabilistic models can be developed.

- The methodology may also be enhanced by applying advanced statistical or machine learning techniques to refine performance evaluation and predictive analysis.

References

- Official Indian Premier League (IPL) Website – T20 League
  Official source for season-wise points tables, match results, and Net Run Rate data used in this study.
  https://www.iplt20.com
- IPL Points Table Archives (2022–2024 Seasons)
  Historical season data accessed through the official IPL website, forming the primary dataset for performance evaluation and ranking analysis.
  https://www.iplt20.com/points-table
- ESPNcricinfo – Cricket Statistics and Analytics
  Used as a secondary reference for understanding performance metrics, Net Run Rate interpretation, and comparative cricket analytics.
  https://www.espncricinfo.com
- James, B. (2001). *The Bill James Historical Baseball Abstract*.
  Referenced conceptually for ideas related to composite indices and performance normalization in sports analytics.